

24/1164**B.C.A. (Fifth Semester)****Examination, 2024****Third Paper****(Computer Networks)***Time : Two Hours]**[Maximum Marks : 75***Note :** Attempt **all** sections as per instructions.**Section-A****(Very Short Answer Type Questions)****Note :** Attempt all the **05 (five)** questions.
Each question carries **02 (two)** marks
and the answer of each question should
not exceed **50** words. $5 \times 2 = 10$

1. (a) Write the names of the different network topologies. Illustrate with a line diagram. 2
- (b) What is the main function of the Data Link layer? 2

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- (c) Distinguish between guided and unguided media with examples. 2
- (d) Name the services available under the ISDN. 2
- (e) What are the network layer services? 2

Section-B**(Short Answer Type Questions)****Note :** Attempt any **05 (five)** questions out of total **08 (Eight)** questions. Each question carries **05 (five)** marks and the answer of each question should not exceed **100** words. $5 \times 5 = 25$

2. (a) What is the function of a Modem? Explain briefly. 5
- (b) State the features of half duplex transmission. 5
- (c) Define connection-oriented service in a Network. Give a few examples. 5
- (d) What is the function of a Router? 5

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- (e) Explain the role of Bridges in networks covering a large geographical area. 5
- (f) What is meant by Shannon Capacity? Explain. 5
- (g) What is meant by a Protocol? Distinguish between character and bit oriented protocols. 5
- (h) What is the role/function of the Sessions Layer? 5

Section-C

(Long Answer Type Questions)

Note : Attempt any **02 (two)** questions out of total **04 (four)** questions. Each question carries **20 (twenty)** marks and the answer of each question should not exceed **400** words. $2 \times 20 = 40$

3. (a) Write a short note on the following, giving neat diagrams: $(6+6+8)$
- (i) Transmission Modes
 - (ii) Distributed Processing
 - (iii) TCP/IP Model

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- (b) Distinguish between the following with examples: $(5 \times 4 = 20)$
 - (i) Alternation and Distortion
 - (ii) TDM and FDM
 - (iii) Synchronous and Asynchronous Protocols
 - (iv) Flow Control and Error Control
- (c) Give an explanatory account of the following: $(7+7+6)$
 - (i) Categories of Networks
 - (ii) Circuit Switching and packet switching
 - (iii) Transport layer functions
- (d) (i) What is congestion control in a Network? Discuss an algorithm to implement congestion control in a Network. $(12+8)$
- (ii) What is Internetworking? Explain.

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